

Linux Practical Examination

Q1. Basic File Operations

- Create a directory named LinuxExam.
- Navigate into the directory.
- Create: o student.txt o course.txt o result.txt
- Display the current working directory.
- Display the contents of the directory.

```
ubuntu@ip-172-31-5-57:~$ echo 'Qustion 1'
Qustion 1
ubuntu@ip-172-31-5-57:~$ mkdir linuxexam
ubuntu@ip-172-31-5-57:~$ cd linuxexam/
ubuntu@ip-172-31-5-57:~/linuxexam$ nano student.txt course.txt result.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ pwd
/home/ubuntu/linuxexam
ubuntu@ip-172-31-5-57:~/linuxexam$ ls
result.txt  student.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ |
```

Q2. File Management

Inside LinuxExam:

- Copy student.txt as student_backup.txt.
- Rename course.txt to linux_course.txt.
- Delete result.txt.
- Create directories: o Documents o Backups o Scripts
- Move student_backup.txt into Backups.
- Display the directory structure.

```
ubuntu@ip-172-31-5-57:~/linuxexam$ echo 'Qustion 2'
Qustion 2
ubuntu@ip-172-31-5-57:~/linuxexam$ cp student.txt student_backup.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ mv course.txt linux_course.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ rm result.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ mkdir document backups scripts
ubuntu@ip-172-31-5-57:~/linuxexam$ mv student_backup.txt backups/
ubuntu@ip-172-31-5-57:~/linuxexam$ ls -R
.:
backups  document  linux_course.txt  scripts  student.txt

./backups:
student_backup.txt

./document:

./scripts:
ubuntu@ip-172-31-5-57:~/linuxexam$ |
```

Q3. File Content Operations

- Add at least 5 lines of student information to student.txt.
- Add at least 5 lines of course information to linux_course.txt.
- Display the contents using cat.
- Display the first 3 lines using head.
- Display the last 2 lines using tail.
- Count the number of lines, words, and characters in student.txt.

```
ubuntu@ip-172-31-5-57:~/linuxexam$ echo 'Qustion 3'
Qustion 3
ubuntu@ip-172-31-5-57:~/linuxexam$ nano student.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ nano linux_course.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ cat student.txt
Student Name: vaibhav patil
Age: 20 years
Course: BCA
Roll Number: 101
City: Pune
ubuntu@ip-172-31-5-57:~/linuxexam$ cat linux_course.txt
Course: Cloud Computing, AWS and DevOps
Topics: Linux, AWS EC2, S3, IAM, VPC, networking and databases.
DevOps: Git, GitHub, CI/CD, Docker, Jenkins and automation.
Practical: Students learn to deploy and manage applications on AWS.
Career: Useful for roles like Cloud Engineer, AWS Engineer and DevOps Engineer.
ubuntu@ip-172-31-5-57:~/linuxexam$ head -3 linux_course.txt
Course: Cloud Computing, AWS and DevOps
Topics: Linux, AWS EC2, S3, IAM, VPC, networking and databases.
DevOps: Git, GitHub, CI/CD, Docker, Jenkins and automation.
ubuntu@ip-172-31-5-57:~/linuxexam$ tail -2 linux_course.txt
Practical: Students learn to deploy and manage applications on AWS.
Career: Useful for roles like Cloud Engineer, AWS Engineer and DevOps Engineer.
ubuntu@ip-172-31-5-57:~/linuxexam$ wc -l linux_course.txt
5 linux_course.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ |
```

Q4. User Management

- Create a user named student01.
- Set a password.
- Verify that the user exists.
- Display the user's UID and GID.
- Display the user's home directory.
- Switch to the student01 account and verify the current username.

```
ubuntu@ip-172-31-5-57:~/linuxexam$ echo 'Qustion 4'
Qustion 4
ubuntu@ip-172-31-5-57:~/linuxexam$ sudo adduser student01
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student01
Enter the new value, or press ENTER for the default
    Full Name []: vaibhav patil
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
ubuntu@ip-172-31-5-57:~/linuxexam$ id student01
uid=1001(student01) gid=1001(student01) groups=1001(student01),100(users)
ubuntu@ip-172-31-5-57:~/linuxexam$ su student01
Password:
student01@ip-172-31-5-57:/home/ubuntu/linuxexam$ pwd
/home/ubuntu/linuxexam
student01@ip-172-31-5-57:/home/ubuntu/linuxexam$ whoim
Command 'whoim' not found, did you mean:
  command 'whois' from deb whois (5.6.6)
  command 'whom' from deb mailutils-mh (1:3.20-3build1)
  command 'whom' from deb nmh (1.8-4)
Try: apt install <deb name>
student01@ip-172-31-5-57:/home/ubuntu/linuxexam$ who
student01@ip-172-31-5-57:/home/ubuntu/linuxexam$ |
```

Q6. File Permissions

Create project.txt and:

- Give owner Read, Write and Execute permission.
- Give Group Read and Execute permission.
- Give Others Read permission only.
- Verify the permissions.

Expected: 754

Then perform the following:

- Change the permission to 640.
- Change the owner of the file to student01.
- Change the group ownership to linuxbatch.
- Verify the final ownership and permissions.

```
ubuntu@ip-172-31-5-57:~/linuxexam$ echo 'Qustion 6'
Qustion 6
ubuntu@ip-172-31-5-57:~/linuxexam$ mkdir project.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ chmod 754 project.txt/
ubuntu@ip-172-31-5-57:~/linuxexam$ ls -l
total 24
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 25 19:29 backups
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 25 19:29 document
-rw-rw-r-- 1 ubuntu ubuntu 312 Aug 25 19:32 linux_course.txt
drwxr-xr-- 2 ubuntu ubuntu 4096 Aug 25 19:39 project.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 25 19:29 scripts
-rw-rw-r-- 1 ubuntu ubuntu 82 Aug 25 19:32 student.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ chmod 640 project.txt/
ubuntu@ip-172-31-5-57:~/linuxexam$ sudo chown student01 project.txt/
ubuntu@ip-172-31-5-57:~/linuxexam$ ls -l
total 24
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 25 19:29 backups
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 25 19:29 document
-rw-rw-r-- 1 ubuntu ubuntu 312 Aug 25 19:32 linux_course.txt
drw-r----- 2 student01 ubuntu 4096 Aug 25 19:39 project.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug 25 19:29 scripts
-rw-rw-r-- 1 ubuntu ubuntu 82 Aug 25 19:32 student.txt
ubuntu@ip-172-31-5-57:~/linuxexam$ |
```

Q7. Permission Challenge

Create:

```
LinuxExam/ ├── public/  
           ├── private/  
           └── shared/
```

Configure permissions so that:

- public can be accessed by everyone.
- private can be accessed only by its owner.
- shared can be accessed by the owner and group members.
- Verify the permissions using `ls -ld`.

```
ubuntu@ip-172-31-5-57:~/linuxexam$ echo 'Qustion 7'  
Qustion 7  
ubuntu@ip-172-31-5-57:~/linuxexam$ mkdir linuxexam  
ubuntu@ip-172-31-5-57:~/linuxexam$ cd linuxexam/  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ mkdir public privet shared  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ ls  
privet public shared  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ chmod 777 public/  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ chmod 700 privet/  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ chmod 770 shared/  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ ls -ld  
drwxrwxr-x 5 ubuntu ubuntu 4096 Aug 25 19:44 .  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ ls -l  
total 12  
drwx----- 2 ubuntu ubuntu 4096 Aug 25 19:44 privet  
drwxrwxrwx 2 ubuntu ubuntu 4096 Aug 25 19:44 public  
drwxrwx--- 2 ubuntu ubuntu 4096 Aug 25 19:44 shared  
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ |
```

Q8. Package Management

- Update the package repository.
- Install Apache Web Server.
- Verify that Apache is installed.
- Display the installed Apache version.
- Start the Apache service.
- Check the status of Apache.
- Configure Apache to start automatically at boot.

```

ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ echo 'Question 8'
Question 8
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ sudo apt-get update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [549 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [127 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5756 B]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [252 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [83.4 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4704 B]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [371 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [71.7 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64v3 Packages [11.4 kB]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [256 B]
Get:27 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:28 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:29 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64v3 Packages [3148 B]
Get:30 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7988 B]
Get:31 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
Get:32 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:33 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:34 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:35 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
Get:36 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Get:37 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [422 kB]
Get:38 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.6 kB]
Get:39 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.8 kB]
Get:40 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:41 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [159 kB]
Get:42 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:43 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:44 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:45 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [357 kB]
Get:46 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [69.0 kB]
Get:47 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64v3 Packages [9028 B]
Get:48 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2660 B]
Get:49 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:50 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Fetched 31.4 MB in 4s (7976 kB/s)
Reading package lists... Done

```

```

ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ sudo apt-get install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap liba
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ld
0 upgraded, 10 newly installed, 0 to remove and 117 not upgraded.
Need to get 2123 kB of archives.
After this operation, 8212 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libapr1t64 amd64 1.7.
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libaprutil1t64 amd64
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libaprutil1-dbd-sqlit
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libaprutil1-ldap amd6
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 liblua5.4-0 amd64 5.4
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2-bin a
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2-data
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2-utils
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2 amd64
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 ssl-cert all 1.1.3ub
Fetched 2123 kB in 0s (48.7 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 85303 files and directories currently installed.)
Preparing to unpack ../0-libapr1t64_1.7.6-3_amd64v3.deb ...
Unpacking libapr1t64:amd64 (1.7.6-3)

```

```

ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ apache2 -v
Server version: Apache/2.4.66 (Ubuntu)
Server built: 2026-07-06T15:33:20
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ sudo service apache2 start
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ sudo service apache2 status
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-25 19:51:24 UTC; 2min 15s ago
 Invocation: c9e1b9dfc073483aa9f2d45a8aa98694
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2844 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 55 (limit: 627)
   Memory: 5.8M (peak: 6.1M)
     CPU: 92ms
   CGroup: /system.slice/apache2.service
           └─2844 /usr/sbin/apache2 -k start -DFOREGROUND
             └─2860 /usr/sbin/apache2 -k start -DFOREGROUND
               └─2861 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 25 19:51:24 ip-172-31-5-57 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 25 19:51:24 ip-172-31-5-57 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-5-57:~/linuxexam/linuxexam$ |

```

Q9. Apache Web Server Configuration

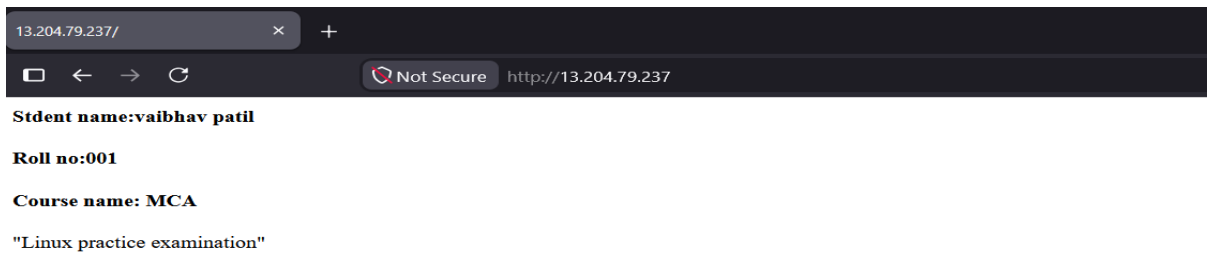
- Find the Apache document root.
- Create a custom index.html.
- Put the following information on the webpage:
 - Student Name
 - Roll Number
 - Course Name
 - "Linux Practical Examination"
- Start/restart Apache.
- Access the webpage using the browser or curl.
- Display the HTTP response using curl.

```

ubuntu@ip-172-31-5-57:~$ echo 'Qustion 9'
Qustion 9
ubuntu@ip-172-31-5-57:~$ cd /var/www/html/
ubuntu@ip-172-31-5-57:/var/www/html$ sudo nano index.html
ubuntu@ip-172-31-5-57:/var/www/html$ sudo service apache2 restart
ubuntu@ip-172-31-5-57:/var/www/html$ curl http://13.204.79.237
<!DOCTYPE html>
<html>
<head>
  <title>Linux Practical Examination</title>
</head>
<body>
  <h1>Linux Practical Examination</h1>

  <p><b>Student Name:</b> vaibhav patil</p>
  <p><b>Roll Number:</b> 101</p>
  <p><b>Course Name:</b> MCA AWS DevOps</p>
</body>
</html>
ubuntu@ip-172-31-5-57:/var/www/html$ |

```



Q11. Search and Text Processing Inside LinuxExam:

- Create a file named students.txt containing at least 10 student records.
- Use grep to find students matching a specified name or pattern.
- Use grep with case-insensitive searching.
- Count matching records.
- Sort the student records.
- Search for a particular file using find.
 - Find all students whose name contains the letter a.
 - or:
 - Find all .txt files inside the LinuxExam directory.

```
ubuntu@ip-172-31-5-57:~$ echo 'Qustion 11'
Qustion 11
ubuntu@ip-172-31-5-57:~$ nano student.txt
ubuntu@ip-172-31-5-57:~$ grep 'pune' student.txt
ubuntu@ip-172-31-5-57:~$ grep 'Pune' student.txt
101 Rahul Sharma BCA Pune
104 Sneha Joshi BCA Pune
107 Akash Pawar BCA Pune
ubuntu@ip-172-31-5-57:~$ grep -i 'pune' student.txt
101 Rahul Sharma BCA Pune
104 Sneha Joshi BCA Pune
107 Akash Pawar BCA Pune
ubuntu@ip-172-31-5-57:~$ grep -c 'Pune' student.txt
3
ubuntu@ip-172-31-5-57:~$ sort student.txt
101 Rahul Sharma BCA Pune
102 Priya Patil BCA Mumbai
103 Amit Kumar BCA Nashik
104 Sneha Joshi BCA Pune
105 Rohan More BCA Jalgaon
106 Neha Singh BCA Nagpur
107 Akash Pawar BCA Pune
108 Pooja Deshmukh BCA Nashik
109 Karan Shah BCA Mumbai
110 Anjali Kulkarni BCA Jalgaon
ubuntu@ip-172-31-5-57:~$ find . -type f -name 'student.txt'
./student.txt
./linuxexam/student.txt
ubuntu@ip-172-31-5-57:~$ cd
ubuntu@ip-172-31-5-57:~$ find . -type f -name "*.txt"
./student.txt
./linuxexam/student.txt
./linuxexam/linux_course.txt
./linuxexam/backups/student_backup.txt
ubuntu@ip-172-31-5-57:~$ |
```